

Unit 3 Test

Gravitational, Electric and Magnetic Fields

K 6 + 12 18 / 18	T 18 18	A 14 10	C 10 10
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SHOW ALL YOUR WORK IN ANSWERING QUESTIONS.

Excellent work!

THINKING

1. What must the separation be between a 4.6 kg object and a 1.7 kg object for their gravitational attraction to have a magnitude of 1.8×10^{-12} N?

Show all your work.

$$m_1 = 4.6 \text{ kg}$$

$$m_2 = 1.7 \text{ kg}$$

$$F_g = 1.8 \times 10^{-12} \text{ N}$$

$$r = ?$$

$$G = 6.67 \times 10^{-11} \text{ N m}^2/\text{kg}^2$$

$$F_g = \frac{G m_1 m_2}{r^2}$$

$$r^2 = \frac{G m_1 m_2}{F_g}$$

$$r = \sqrt{\frac{G m_1 m_2}{F_g}}$$

$$r = \sqrt{\frac{6.67 \times 10^{-11} \cdot 4.6 \cdot 1.7}{1.8 \times 10^{-12}}}$$

$$r = 17.02 \text{ m}$$

$$\therefore r = 17 \text{ m apart}$$

Final answer: 17 m apart

[3 marks]

2. A satellite is in circular orbits around the earth. The orbit for the satellite is at a height of 720 km above the earth's surface. Determine the orbital speed for the satellite.

Show all your work.

$$V = ?$$

$$r = r_E + r_s$$

$$r = 6.38 \times 10^6 + 720,000 \text{ m} = 7.1 \times 10^6 \text{ m}$$

$$M_E = 5.98 \times 10^{24} \text{ kg}$$

$$G = 6.67 \times 10^{-11} \text{ N m}^2/\text{kg}^2$$

$$V_c = \sqrt{\frac{GM}{r}}$$

$$V_c = \sqrt{\frac{6.67 \times 10^{-11} \cdot 5.98 \times 10^{24}}{7.1 \times 10^6}}$$

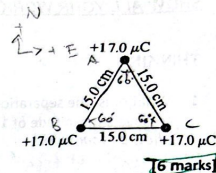
$$V_c = 7495.22 \text{ m/s}$$

$$\therefore V_c = 7.5 \times 10^3 \text{ m/s}$$

Final answer: $7.5 \times 10^3 \text{ m/s}$

[3 marks]

3. Three positive particles of equal charge, $+17.0 \times 10^{-6} \text{ C}$, are located at the corners of an equilateral triangle of side 15.0 cm as shown in the diagram. Calculate the magnitude and direction of the net force on each particle due to the other two.
Show all your work.



Because the particles are identical relative to the other two, the magnitudes of net force will be the same, while the direction will.

Calculate magnitude of ABC on A

$$F_{AB} = \frac{k|q_1||q_2|}{r^2}$$

$$F_{AB} = \frac{k|q_1||q_2|}{r^2}$$

AB:

$$k = 9 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$$

$$q_1 = 17 \mu\text{C} = 17 \times 10^{-6} \text{ C}$$

$$q_2 = 17 \mu\text{C} = 17 \times 10^{-6} \text{ C}$$

$$r = 0.15 \text{ m}$$

$$F_{AB} = \frac{9 \times 10^9 \cdot 17 \times 10^{-6} \cdot 17 \times 10^{-6}}{0.15^2}$$

$$F_{AB} = 115.6 \text{ N}$$

$$k = 9 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$$

$$q_1 = 17 \times 10^{-6} \text{ C}$$

$$q_2 = 17 \times 10^{-6} \text{ C}$$

$$r = 0.15 \text{ m}$$

$$F_{AC} = \frac{9 \times 10^9 \cdot 17 \times 10^{-6} \cdot 17 \times 10^{-6}}{0.15^2}$$

$$F_{AC} = 115.6 \text{ N}$$



$$\cos 60^\circ = \frac{F_{x1}}{115.6 \text{ N}}$$

$$F_{x1} = 115.6 \text{ N} \cos 60^\circ$$

$$F_{x1} = 57.8 \text{ N [E]}$$

$$F_{y1} = 115.6 \text{ N} \sin 60^\circ$$

$$F_{y1} = 100.11 \text{ N [U]}$$

$$c^2 = a^2 + b^2$$

$$F_{\text{net A}} = (F_{x1} + F_{x2}) + (F_{y1} + F_{y2})$$

$$F_{\text{net A}} = \sqrt{(57.8 \text{ N} + 57.8 \text{ N})^2 + (100.11 \text{ N} + 100.11 \text{ N})^2}$$

$$\therefore F_{\text{net A}} = 200.22 \text{ N [N]}$$



$$F_{x2} = 115.6 \text{ N} \cos 60^\circ$$

$$F_{x2} = 57.8 \text{ N [W]}$$

$$F_{y2} = 115.6 \text{ N} \sin 60^\circ$$

$$F_{y2} = 100.11 \text{ N [U]}$$

magnitudes of force are the same, but direction (s) direction is perpendicular to the other two charges.

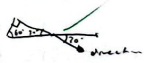
For B:

For C:



$$\therefore 200 \text{ N [W } 30^\circ \text{ S]}$$

For B



$$\therefore 200 \text{ N [E } 30^\circ \text{ S]}$$

$$\therefore 200 \text{ N [E } 30^\circ \text{ S]}$$

For C

$$\text{Final answer: } F_{\text{net B}} = 200 \text{ N [N]}$$

$$\text{Final answer: } F_{\text{net B}} = 200 \text{ N [W } 30^\circ \text{ S]}$$

$$\text{Final answer: } F_{\text{net C}} = 200 \text{ N [E } 30^\circ \text{ S]}$$

4. An electron and a proton are initially very far apart (effectively an infinite distance apart). They are then brought together to form a hydrogen atom, in which the electron orbits the proton at an average distance of $5.29 \times 10^{-11} \text{ m}$. What is the change in the electric potential energy (ΔE_E)?

Show all your work.

[3 marks]

$$E_{E\infty} = 0, \text{ Because infinite distance - has no } E_E$$

$$E_E = k \frac{q_1 q_2}{r}$$

$$k = 9 \times 10^9 \text{ Nm}^2/\text{C}^2$$

$$\text{proton: } q_1 = 1.6 \times 10^{-19} \text{ C}$$

$$\text{electron: } q_2 = -1.6 \times 10^{-19} \text{ C}$$

$$r = 5.29 \times 10^{-11} \text{ m}$$

$$E_E = \frac{9 \times 10^9 \cdot 1.6 \times 10^{-19} \cdot (-1.6 \times 10^{-19})}{5.29 \times 10^{-11} \text{ m}}$$

$$E_{E\infty} = -41.36 \times 10^{-18} \text{ Jm}$$

$$\Delta E_E = E_{E\infty} - E_E$$

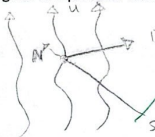
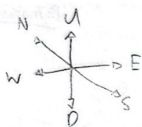
$$\Delta E_E = -4.36 \times 10^{-18} \text{ Nm} \rightarrow 0$$

$$\therefore \Delta E_E = -4.36 \times 10^{-18} \text{ Nm}$$

$$\text{Final answer: } -4.36 \times 10^{-18} \text{ Nm}$$

5. In a 1.25 T magnetic field directed vertically upward, a particle having a charge of magnitude $8.50 \times 10^{-3} \text{ C}$ and initially moving northward at 4.75 km/s is deflected toward the east.

a) What is the sign of the charge of this particle? Make a sketch to illustrate how you found your answer.



Right hand rule
 • thumb forward (fourth)
 • fingers up
 • palm east
 follows deflection therefore positive.

[1 mark]

Final answer: Positive

b) Calculate the magnetic force on the particle. Show all your work.

$$B = 1.25 \text{ T}$$

$$v = 4.75 \text{ km/s} = 4750 \text{ m/s [N]}$$

$$\theta = 90^\circ$$

$$q = 8.5 \times 10^{-3} \text{ C}$$

$$F = |q| v B \sin \theta$$

$$F = |8.5 \times 10^{-3}| \cdot 4750 \cdot 1.25 \cdot \sin 90^\circ$$

$$\therefore F = 50.47 \text{ N [E]}$$

[2 marks]

Final answer: 50.5 N [E]

Unit 3 Test

Gravitational, Electric and Magnetic Fields

K	T	A	C
/18	/18	9/10	10/10

SHOW ALL YOUR WORK IN ANSWERING QUESTIONS.

APPLICATION

1. The drawing shows a situation in which charges are placed on the x and y axes. They are all located at the same distance of 6.1 cm from the origin O. Determine the magnitude and the direction of the net electric field at the origin. *Show all your work.*

$$E = k \frac{q}{r^2}$$

$$r_A = 6.1 \text{ cm} = 0.061 \text{ m}$$

$$r_B = 6.1 \text{ cm} = 0.061 \text{ m}$$

$$r_C = 6.1 \text{ cm} = 0.061 \text{ m}$$

$$k = 8.99 \times 10^9 \text{ Nm}^2/\text{C}^2$$

$$q_A = -50 \mu\text{C} = -5 \times 10^{-6} \text{ C}$$

$$q_B = -30 \mu\text{C} = -3 \times 10^{-6} \text{ C}$$

$$q_C = 2 \mu\text{C} = 2 \times 10^{-6} \text{ C}$$

$$A:$$

$$E_A = k \frac{q_A}{r_A^2}$$

$$E_A = \frac{8.99 \times 10^9 \cdot (-5 \times 10^{-6})}{0.061^2}$$

$$E_A = -1.21 \times 10^7 \text{ N/C}$$

$$B:$$

$$E_B = k \frac{q_B}{r_B^2}$$

$$E_B = \frac{8.99 \times 10^9 \cdot (-3 \times 10^{-6})}{0.061^2}$$

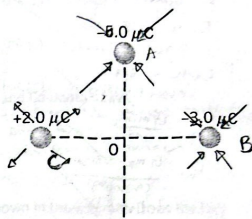
$$E_B = -7.25 \times 10^6 \text{ N/C}$$

$$C:$$

$$E_C = k \frac{q_C}{r_C^2}$$

$$E_C = \frac{8.99 \times 10^9 \cdot (2 \times 10^{-6})}{0.061^2}$$

$$E_C = 4.83 \times 10^6 \text{ N/C}$$



$$E_A = -1.21 \times 10^7 \text{ N/C} \text{ along y}$$

$$E_B = -7.25 \times 10^6 \text{ N/C} \text{ along x}$$

$$E_C = 4.83 \times 10^6 \text{ N/C} \text{ along x}$$

[4 marks]

$$E_x = E_B + E_C$$

$$E_y = E_A$$

$$E_x = 1.208 \times 10^7 \text{ N/C} [E]$$

$$E_y = -1.21 \times 10^7 \text{ N/C} [N]$$

$$E_{\text{net}}^2 = E_x^2 + E_y^2$$

$$E_{\text{net}} = \sqrt{(1.208 \times 10^7)^2 + (-1.21 \times 10^7)^2}$$

$$E_{\text{net}} = 1.71 \times 10^7 \text{ N/C}$$

$$\theta = \tan^{-1} \left(\frac{E_y}{E_x} \right)$$

$$\theta = \tan^{-1} \left(\frac{-1.21 \times 10^7}{1.208 \times 10^7} \right)$$

$$\theta = 15^\circ \text{ N}$$

$$E_{\text{net}} = 1.71 \times 10^7 \text{ N/C} [E \text{ and } N]$$

Final answer: $E_{\text{net}} = 1.71 \times 10^7 \text{ N/C} [E \text{ and } N]$

2. Every few hundred years most of the planets line up on the same side of the Sun as shown in the diagram. Calculate the total force on the Earth due to Venus, Jupiter, and Saturn, assuming all four planets are in a line. The mean distances of the four planets from the Sun are 108 million km, 150 million km, 778 million km, and 1430 million km. The masses are as follows:

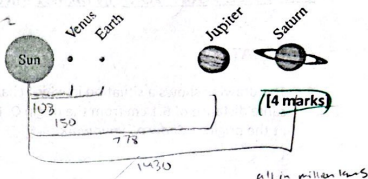
$$m_{\text{Venus}} = 0.815 m_{\text{Earth}}$$

$$m_{\text{Jupiter}} = 318 m_{\text{Earth}}$$

$$m_{\text{Saturn}} = 95.1 m_{\text{Earth}}$$

Show all your work.

$$G = 6.67 \times 10^{-11} \text{ N m}^2/\text{kg}^2$$



$$\text{Earth to Venus} = 150 \text{ mil} - 108 \text{ mil} = 42 \text{ mil km} = 4.2 \times 10^{10} \text{ m}$$

$$\text{Earth to Jupiter} = 778 \text{ mil} - 150 \text{ mil} = 628 \text{ mil km} = 6.28 \times 10^{11} \text{ m}$$

$$\text{Earth to Saturn} = 1430 \text{ mil} - 150 \text{ mil} = 1280 \text{ mil km} = 1.28 \times 10^{12} \text{ m}$$

$$F_g = \frac{G m_1 m_2}{r^2}$$

$$F_v = \frac{G m_E \cdot 0.815 m_E}{(4.2 \times 10^{10})^2}$$

$$F_v = 1.1 \times 10^{18} \text{ N}$$

$$F_j = \frac{G m_E \cdot 318 m_E}{(6.28 \times 10^{11})^2}$$

$$F_j = 1.92 \times 10^{18} \text{ N}$$

$$F_s = \frac{G m_E \cdot 95.1 m_E}{(1.28 \times 10^{12})^2}$$

$$F_s = 1.38 \times 10^{17} \text{ N}$$

$$\text{net force} = F_v + F_j + F_s$$

$$\text{total force} = F_v + F_j + F_s$$

$$\text{net force} = 1.92 \times 10^{18} + 1.38 \times 10^{17} + 1.1 \times 10^{18}$$

$$\text{net force} = 9.58 \times 10^{17} \text{ [R]}$$

assuming you want net force and not total sum?

$$\text{Final answer: } 9.58 \times 10^{17} \text{ [R]}$$

3. A positively charged particle in a nonuniform magnetic field follows the trajectory as shown in the diagram. Indicate the direction of the magnetic field at points a, b, c and e near the path, assuming the path is always in the plane of the page. Explain your answers.

no field

- i) The direction of the magnetic field at point a: right/left line

- ii) The direction of the magnetic field at point b: into the page

- iii) The direction of the magnetic field at point c: out of the page

- iv) The direction of the magnetic field at point e: into the page

Explanation

- i) at point A, no force is being applied to change the trajectory of the charge, meaning there is no field or it is parallel to the charge. [2 marks]
- ii) at point B, a force is being applied up to the charge, meaning the field would point into the page.
- iii) at point C, a force is being applied down to the charge, meaning the field would point out of the page.
- iv) at point E, a force is being applied up to the charge, meaning the field is pointing into the page.

COMMUNICATION

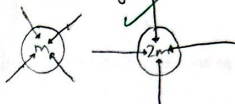
1. If the mass of Earth were much greater than its actual value, but its radius was the same, what would be the effect on the following:

a) Earth's surface gravitational field strength.

b) The evolution of human bone structure?

a) due to the equation $g = \frac{GM}{r^2}$, if the mass of earth (M) increased, the force of gravity would increase proportionally. This means that the overall gravitational field strength would be higher. (2 marks)

b) Due to a higher gravitational field strength, humans would develop a stronger bone structure to hold us up under their higher field strength. Our legs support us under the force gravity exerts on our masses. (2 marks)

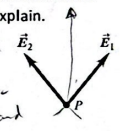


2. Given Universal Law of Gravitation and Coulomb's Law:

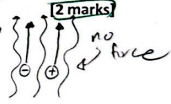
$$F_G = \frac{Gm_1m_2}{r_{12}^2} \quad \text{and} \quad F_E = \frac{kq_1q_2}{r_{12}^2}$$

State two distinct similarities between the Universal Law of Gravitation and Coulomb's Law.

1. Both laws' forces are inversely proportional to the change in distance between two masses / charges, meaning changing the distance will affect both forces by the same factor. (2 marks)
2. Both laws have a constant which calculates for the force acting on a mass / particle in between two masses or charges. (2 marks)
3. The electric fields at point P due to the positive charges q_1 and q_2 are shown in the diagram. Does the fact that $\vec{E}_1$ and $\vec{E}_2$ cross each other violate the statement that electric field lines never cross? Explain. (2 marks)
- When $\vec{E}_1$ & $\vec{E}_2$ intersect at P, it doesn't violate the principle that field lines do not cross. The vectors $\vec{E}_1$ and $\vec{E}_2$ are separate components which will result in a resultant field at P. Therefore no electric field lines cross. The direction of the resultant vector for the electric field is unique. The resultant field will point up, between $\vec{E}_1$ and $\vec{E}_2$, depending on each of their magnitudes.



4. Can a charged particle move through a magnetic field without experiencing any magnetic force? If so, how? If not, why not? (2 marks)
- Yes it can. If the charged particle moves parallel through a magnetic field, it will not experience a magnetic force. The magnetic force is calculated with $F = qv \sin \theta$, and if $\theta = 0$, $\sin \theta = 0$, and F would be zero. (2 marks)



Communication marks are awarded throughout the test according to the shown rubric.

[2 marks]

	Level 1	Level 2	Level 3	Level 4
Correct use of mathematical/ physical symbols, labels, and conventions (e.g. units, proper component form for the values, positive direction given, etc...)	Few mathematical/ physical conventions are used correctly. [1/2]	Some mathematical/ physical conventions are used correctly. [1]	Most mathematical/ physical conventions are used correctly. [1-1/2]	Consistently uses mathematical/ physical conventions correctly. [2]